Chapter 10
Fitting Regression Models
Solutions

10.1S. The tensile strength of a paper product is related to the amount of hardwood in the pulp. Ten samples are produced in the pilot plant, and the data obtained are shown in the following table.

Strength	Percent Hardwood	Strength	Percent Hardwood
160	10	181	20
171	15	188	25
175	15	193	25
182	20	195	28
184	20	200	30

(a) Fit a linear regression model relating strength to percent hardwood.

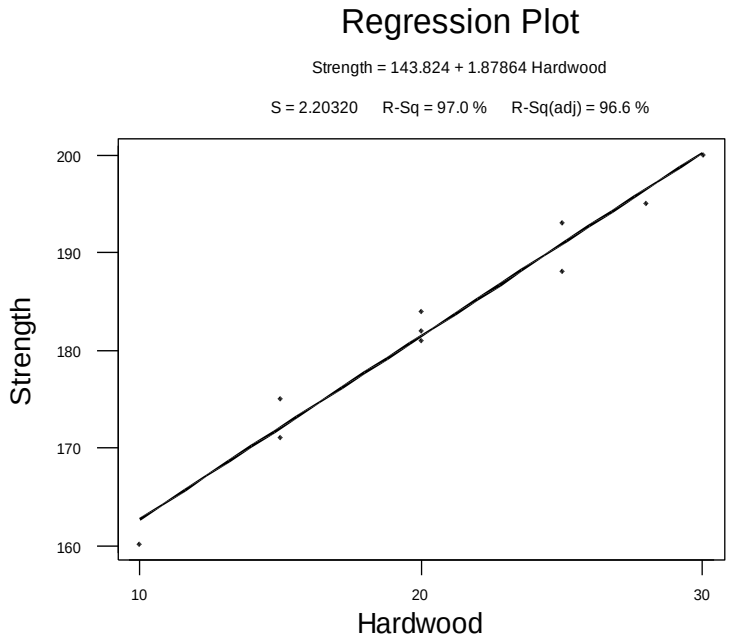
Minitab Output

Regression Analysis: Strength versus Hardwood

```
The regression equation is
Strength = 144 + 1.88 Hardwood

Predictor        Coef     SE Coef          T        P
Constant      143.824       2.522      57.04    0.000
Hardwood       1.8786      0.1165      16.12    0.000

S = 2.203              R-Sq = 97.0%         R-Sq(adj) = 96.6%
PRESS = 66.2665        R-Sq(pred) = 94.91%
```



(b) Test the model in part (a) for significance of regression.

Minitab Output

```
Analysis of Variance

Source            DF          SS          MS         F         P
Regression         1      1262.1      1262.1    260.00     0.000
Residual Error     8        38.8         4.9
  Lack of Fit      4        13.7         3.4      0.54     0.716
  Pure Error       4        25.2         6.3
Total              9      1300.9

3 rows with no replicates

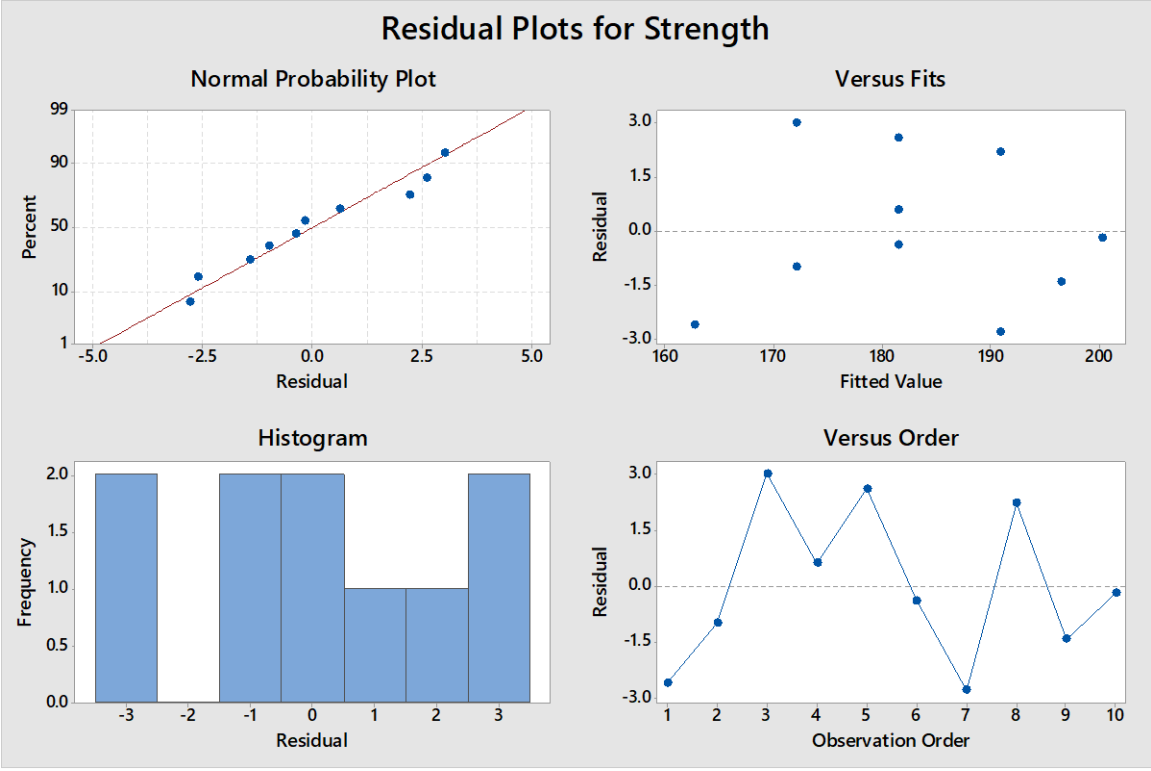
No evidence of lack of fit (P > 0.1)
```

(c) Find a 95 percent confidence interval on the parameter β_1.

The 95 percent confidence interval is:

$$\hat{\beta}_1 - t_{\alpha/2,n-p}\,se\left(\hat{\beta}_1\right) \le \beta_1 \le \hat{\beta}_1 + t_{\alpha/2,n-p}\,se\left(\hat{\beta}_1\right)$$
$$1.8786 - 2.3060\left(0.1165\right) \le \beta_1 \le 1.8786 + 2.3060\left(0.1165\right)$$
$$1.6900 \le \beta_1 \le 2.1473$$

10.2. Plot the residuals from Problem 10.1 and comment on model adequacy.



There is nothing unusual about the residual plots. The underlying assumptions have been met.

10.3. Using the results of Problem 10.1, test the regression model for lack of fit.

The Minitab output below identifies no evidence of lack of fit.

Minitab Output

```
Analysis of Variance

Source            DF          SS          MS         F        P
Regression         1      1262.1      1262.1    260.00    0.000
Residual Error     8        38.8         4.9
  Lack of Fit      4        13.7         3.4      0.54    0.716
  Pure Error       4        25.2         6.3
Total              9      1300.9

3 rows with no replicates

No evidence of lack of fit (P > 0.1)
```

10.4. A study was performed on wear of a bearing y and its relationship to x_1 = oil viscosity and x_2 = load. The following data were obtained.

y	x_1	x_2
193	1.6	851
230	15.5	816
172	22.0	1058
91	43.0	1201
113	33.0	1357
125	40.0	1115

(a) Fit a multiple linear regression model to the data.

Minitab Output

Regression Analysis: Wear versus Viscosity, Load

```
The regression equation is
Wear = 351 - 1.27 Viscosity - 0.154 Load

Predictor        Coef     SE Coef          T        P       VIF
Constant       350.99       74.75       4.70    0.018
Viscosit       -1.272       1.169      -1.09    0.356       2.6
Load         -0.15390     0.08953      -1.72    0.184       2.6

S = 25.50              R-Sq = 86.2%         R-Sq(adj) = 77.0%
PRESS = 12696.7        R-Sq(pred) = 10.03%
```

(b) Test for significance of regression.

Minitab Output

```
Analysis of Variance

Source            DF          SS          MS         F        P
Regression         2     12161.6      6080.8      9.35    0.051
Residual Error     3      1950.4       650.1
Total              5     14112.0

No replicates. Cannot do pure error test.

Source       DF      Seq SS
Viscosit      1     10240.4
Load          1      1921.2

* Not enough data for lack of fit test
```

(c) Compute *t* statistics for each model parameter. What conclusions can you draw?

Minitab Output

Regression Analysis: Wear versus Viscosity, Load

```
The regression equation is
Wear = 351 - 1.27 Viscosity - 0.154 Load

Predictor        Coef     SE Coef          T        P       VIF
Constant       350.99       74.75       4.70    0.018
Viscosit       -1.272       1.169      -1.09    0.356       2.6
Load         -0.15390     0.08953      -1.72    0.184       2.6

S = 25.50              R-Sq = 86.2%         R-Sq(adj) = 77.0%
PRESS = 12696.7        R-Sq(pred) = 10.03%
```

The *t*-tests are shown in part (a). Notice that overall regression is significant (part(b)), but neither variable has a large *t*-statistic. This could be an indicator that the regressors are nearly linearly dependent.

10.5S. The brake horsepower developed by an automobile engine on a dynamometer is thought to be a function of the engine speed in revolutions per minute (rpm), the road octane number of the fuel, and the engine compression. An experiment is run in the laboratory and the data that follow are collected.

Brake Horsepower	rpm	Road Octane Number	Compression
225	2000	90	100
212	1800	94	95
229	2400	88	110
222	1900	91	96
219	1600	86	100
278	2500	96	110
246	3000	94	98
237	3200	90	100
233	2800	88	105
224	3400	86	97
223	1800	90	100
230	2500	89	104

(a) Fit a multiple linear regression model to the data.

Minitab Output

Regression Analysis: Horsepower versus rpm, Octane, Compression

```
The regression equation is
Horsepower = - 266 + 0.0107 rpm + 3.13 Octane + 1.87 Compression

Predictor        Coef     SE Coef          T        P       VIF
Constant      -266.03       92.67      -2.87    0.021
rpm          0.010713    0.004483       2.39    0.044       1.0
Octane         3.1348      0.8444       3.71    0.006       1.0
Compress       1.8674      0.5345       3.49    0.008       1.0

S = 8.812              R-Sq = 80.7%         R-Sq(adj) = 73.4%
PRESS = 2494.05        R-Sq(pred) = 22.33%
```

(b) Test for significance of regression. What conclusions can you draw?

Minitab Output

```
Analysis of Variance

Source            DF          SS          MS         F        P
Regression         3     2589.73      863.24     11.12    0.003
Residual Error     8      621.27       77.66
Total             11     3211.00
r No replicates. Cannot do pure error test.

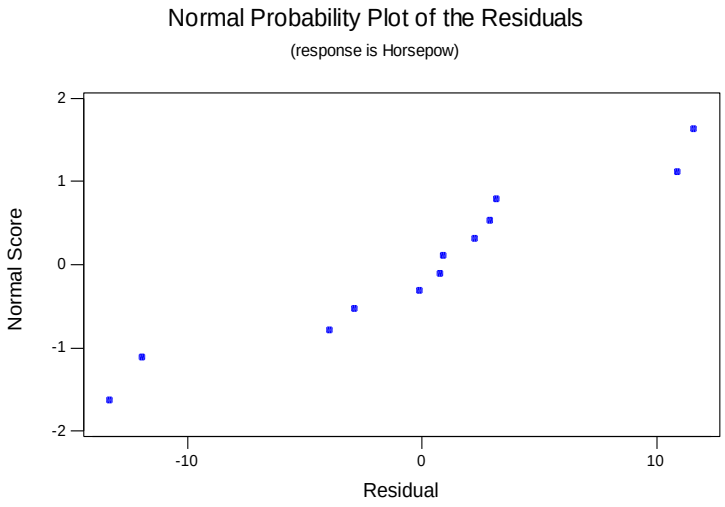
Source       DF      Seq SS
rpm           1      509.35
Octane        1     1132.56
Compress      1      947.83

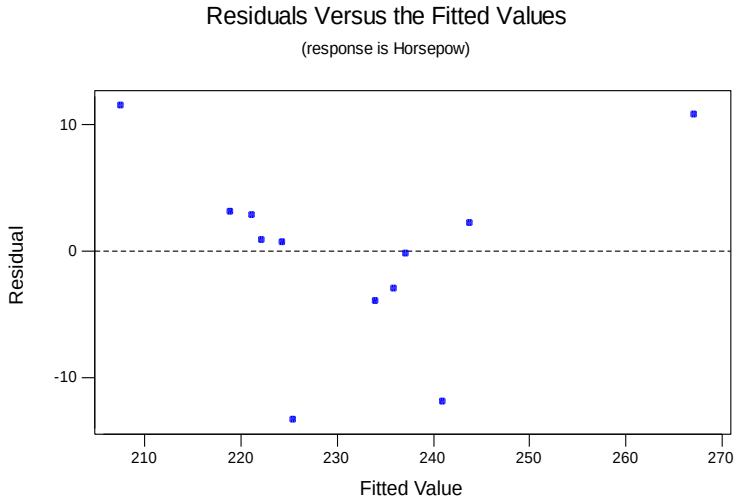
Lack of fit test
Possible interactions with variable Octane (P-Value = 0.028)
Possible lack of fit at outer X-values       (P-Value = 0.000)
Overall lack of fit test is significant at P = 0.000
```

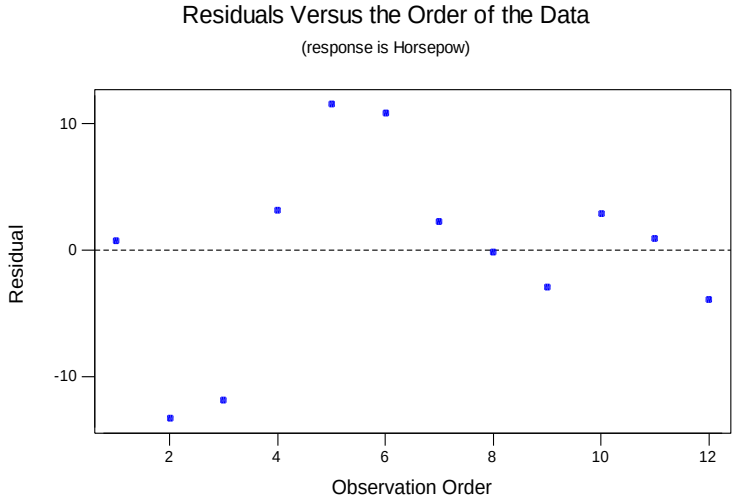
(c) Based on *t* tests, do you need all three regressor variables in the model?

Yes, all the regressor variables are important.

10.6. Analyze the residuals from the regression model in Problem 10.5. Comment on model adequacy.







The normal probability plot is satisfactory, as is the plot of residuals versus run order (assuming that observation order is run order). The plot of residuals versus predicted exhibits a slight "bow" shape. This could be an indication of lack of fit. It might be useful to add some interaction terms to the model.

10.7S. The yield of a chemical process is related to the concentration of the reactant and the operating temperature. An experiment has been conducted with the following results.

Yield	Concentration	Temperature
81	1.00	150
89	1.00	180
83	2.00	150
91	2.00	180
79	1.00	150
87	1.00	180
84	2.00	150
90	2.00	180

(a) Suppose we wish to fit a main effects model to this data. Set up the **X'X** matrix using the data exactly as it appears in the table.

$$\begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1.00 & 1.00 & 2.00 & 2.00 & 1.00 & 1.00 & 2.00 & 2.00 \\ 150 & 180 & 150 & 180 & 150 & 180 & 150 & 180 \end{bmatrix} \begin{bmatrix} 1 & 1.00 & 150 \\ 1 & 1.00 & 180 \\ 1 & 2.00 & 150 \\ 1 & 2.00 & 180 \\ 1 & 1.00 & 150 \\ 1 & 1.00 & 180 \\ 1 & 2.00 & 150 \\ 1 & 2.00 & 180 \end{bmatrix} = \begin{bmatrix} 8 & 12 & 1320 \\ 12 & 20 & 1980 \\ 1320 & 1980 & 219600 \end{bmatrix}$$

(b) Is the matrix you obtained in part (a) diagonal? Discuss your response.

The **X'X** is not diagonal, even though an orthogonal design has been used. The reason is that we have worked with the natural factor levels, not the orthogonally coded variables.

(c) Suppose we write our model in terms of the "usual" coded variables

$$x_1 = \frac{Conc - 1.5}{0.5}, \; x_2 = \frac{Temp - 165}{15}$$

Set up the **X'X** matrix for the model in terms of these coded variables. Is this matrix diagonal? Discuss your response.

$$\begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ -1 & -1 & 1 & 1 & -1 & -1 & 1 & 1 \\ -1 & 1 & -1 & 1 & -1 & 1 & -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & -1 & -1 \\ 1 & -1 & 1 \\ 1 & 1 & -1 \\ 1 & 1 & 1 \\ 1 & -1 & -1 \\ 1 & -1 & 1 \\ 1 & 1 & -1 \\ 1 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 8 & 0 & 0 \\ 0 & 8 & 0 \\ 0 & 0 & 8 \end{bmatrix}$$

The **X'X** matrix is diagonal because we have used the orthogonally coded variables.

(d) Define a new set of coded variables

$$x_1 = \frac{Conc - 1.0}{1.0}, \; x_2 = \frac{Temp - 150}{30}$$

Set up the **X'X** matrix for the model in terms of this set of coded variables. Is this matrix diagonal? Discuss your response.

$$\begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 0 & 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \\ 1 & 0 & 0 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 8 & 4 & 4 \\ 4 & 4 & 2 \\ 4 & 2 & 4 \end{bmatrix}$$

The **X'X** is not diagonal, even though an orthogonal design has been used. The reason is that we have not used orthogonally coded variables.

(e) Summarize what you have learned from this problem about coding the variables.

If the design is orthogonal, use the orthogonal coding. This not only makes the analysis somewhat easier, but it also results in model coefficients that are easier to interpret because they are both dimensionless and uncorrelated.

10.8. Consider the 2^4 factorial experiment in Example 6.2. Suppose that the last two observations are missing. Reanalyze the data and draw conclusions. How do these conclusions compare with those from the original example?

The regression analysis with two data points missing indicates that the same effects are important.

Minitab Output

Regression Analysis: Rate versus A, B, C, D, AB, AC, AD, BC, BD, CD

```
The regression equation is
Rate = 71.4 + 10.1 A + 2.87 B + 6.25 C + 8.62 D - 0.66 AB - 9.78 AC + 7.59 AD
           + 2.50 BC + 1.12 BD + 0.75 CD

Predictor        Coef     SE Coef          T        P       VIF
Constant       71.375       1.673      42.66    0.000
A              10.094       1.323       7.63    0.005       1.1
B               2.875       1.673       1.72    0.184       1.7
C               6.250       1.673       3.74    0.033       1.7
D               8.625       1.673       5.15    0.014       1.7
AB             -0.656       1.323      -0.50    0.654       1.1
AC             -9.781       1.323      -7.39    0.005       1.1
AD              7.594       1.323       5.74    0.010       1.1
BC              2.500       1.673       1.49    0.232       1.7
BD              1.125       1.673       0.67    0.549       1.7
CD              0.750       1.673       0.45    0.684       1.7

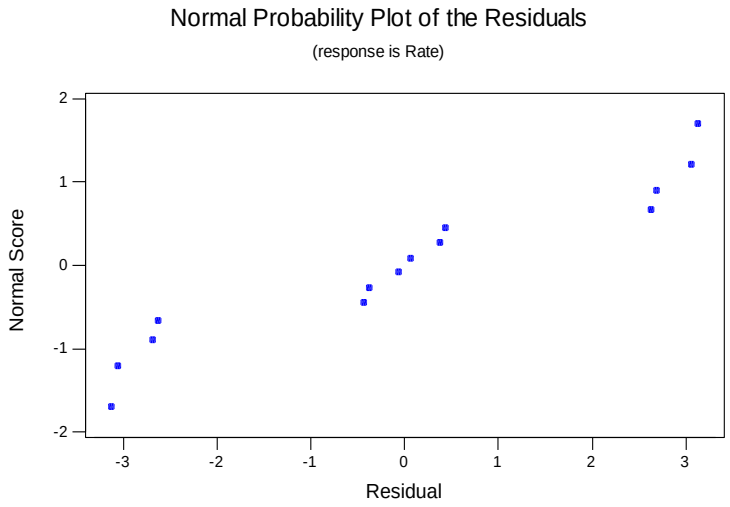
S = 4.732              R-Sq = 98.7%         R-Sq(adj) = 94.2%
PRESS = 1493.06        R-Sq(pred) = 70.20%

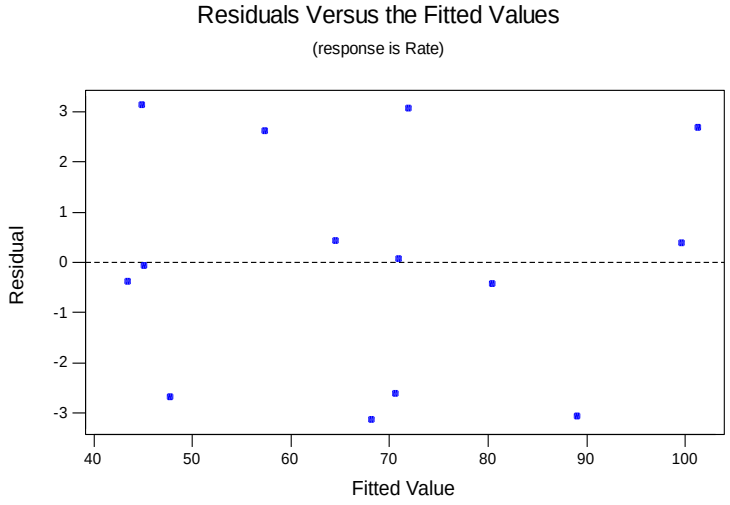
Analysis of Variance

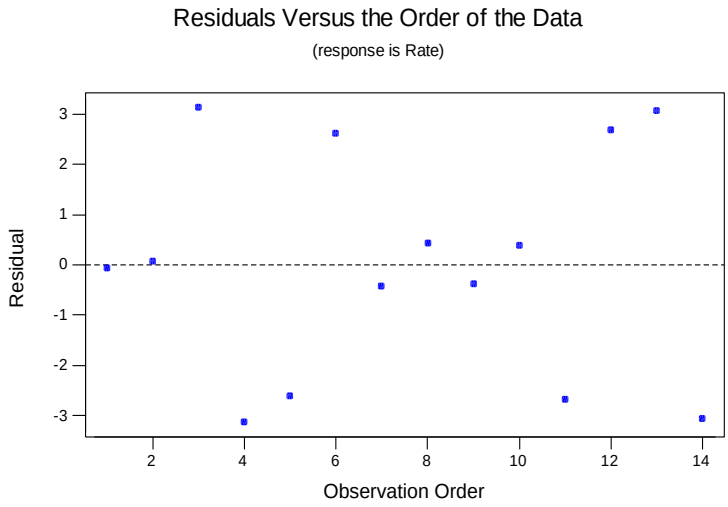
Source            DF          SS          MS         F        P
Regression        10     4943.17      494.32     22.07    0.014
Residual Error     3       67.19       22.40
Total             13     5010.36

No replicates. Cannot do pure error test.
```

Source	DF	Seq SS
A	1	1543.50
B	1	1.52
C	1	177.63
D	1	726.01
AB	1	1.17
AC	1	1702.53
AD	1	738.11
BC	1	42.19
BD	1	6.00
CD	1	4.50







The residual plots are acceptable; therefore, the underlying assumptions are valid.

10.9. Given the following data, fit the second-order polynomial regression model

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_{11} x_1^2 + \beta_{22} x_2^2 + \beta_{12} x_1 x_2 + \varepsilon$$

y	x_1	x_2
26	1.0	1.0
24	1.0	1.0
175	1.5	4.0
160	1.5	4.0
163	1.5	4.0
55	0.5	2.0
62	1.5	2.0
100	0.5	3.0
26	1.0	1.5
30	0.5	1.5
70	1.0	2.5
71	0.5	2.5

After you have fit the model, test for significance of regression.

Minitab Output

Regression Analysis: y versus x1, x2, x1^2, x2^2, x1x2

```
The regression equation is
y = 24.4 - 38.0 x1 + 0.7 x2 + 35.0 x1^2 + 11.1 x2^2 - 9.99 x1x2

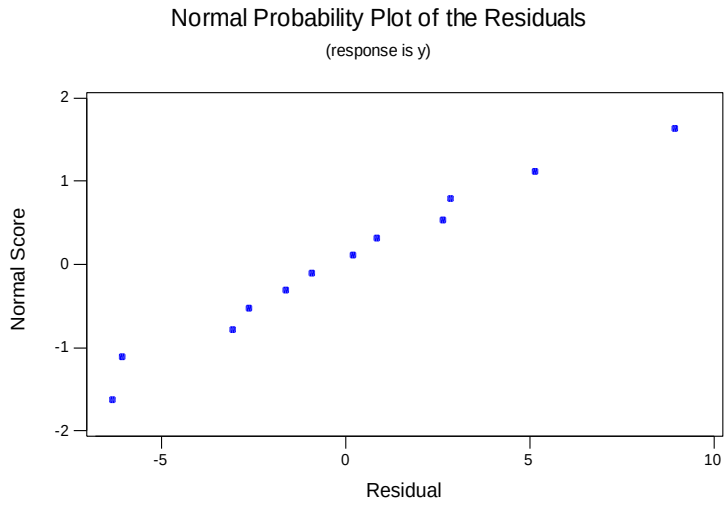
Predictor        Coef     SE Coef          T        P       VIF
Constant        24.41       26.59       0.92    0.394
x1             -38.03       40.45      -0.94    0.383      89.6
x2               0.72       11.69       0.06    0.953      52.1
x1^2            34.98       21.56       1.62    0.156     103.9
x2^2           11.066       3.158       3.50    0.013     104.7
x1x2           -9.986       8.742      -1.14    0.297     105.1

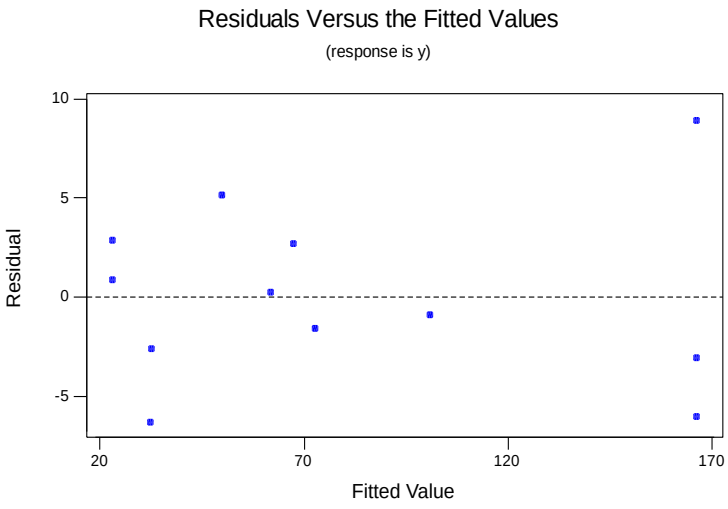
S = 6.042              R-Sq = 99.4%         R-Sq(adj) = 98.9%
PRESS = 1327.71        R-Sq(pred) = 96.24%
r Analysis of Variance
```

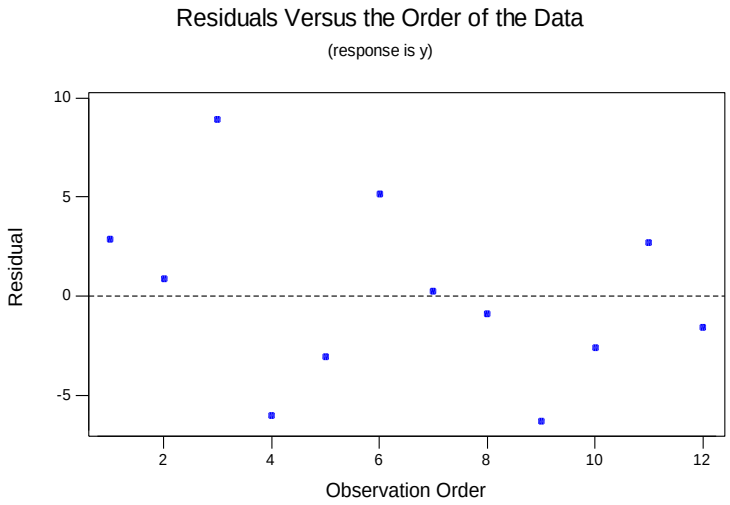
```
Source            DF          SS          MS         F        P
Regression         5     35092.6      7018.5    192.23    0.000
Residual Error     6       219.1        36.5
  Lack of Fit      3        91.1        30.4      0.71    0.607
  Pure Error       3       128.0        42.7
Total             11     35311.7

7 rows with no replicates

Source       DF      Seq SS
x1            1     11552.0
x2            1     22950.3
x1^2          1        21.9
x2^2          1       520.8
x1x2          1        47.6
```







10.10.

(a) Consider the quadratic regression model from Problem 10.9. Compute *t* statistics for each model parameter and comment on the conclusions that follow from the quantities.

Minitab Output

Predictor	Coef	SE Coef	T	P	VIF
Constant	24.41	26.59	0.92	0.394	
x1	-38.03	40.45	-0.94	0.383	89.6
x2	0.72	11.69	0.06	0.953	52.1
x1^2	34.98	21.56	1.62	0.156	103.9
x2^2	11.066	3.158	3.50	0.013	104.7
x1x2	-9.986	8.742	-1.14	0.297	105.1

x_2^2 is the only model parameter that is statistically significant with a *t*-value of 3.50. A logical model might also include x_2 to preserve model hierarchy.

(b) Use the extra sum of squares method to evaluate the value of the quadratic terms, x_1^2, x_2^2 and x_1x_2 to the model.

The extra sum of squares due to $\boldsymbol{\beta}_2$ is

$$SS_R\left(\boldsymbol{\beta}_2\left|\boldsymbol{\beta}_{0,}\boldsymbol{\beta}_1\right.\right)=SS_R\left(\boldsymbol{\beta}_0,\boldsymbol{\beta}_1,\boldsymbol{\beta}_2\right)-SS_R\left(\boldsymbol{\beta}_0,\boldsymbol{\beta}_1\right)=SS_R\left(\boldsymbol{\beta}_1,\boldsymbol{\beta}_2\left|\boldsymbol{\beta}_0\right.\right)-SS_R\left(\boldsymbol{\beta}_1\left|\boldsymbol{\beta}_0\right.\right)$$

$SS_R\left(\boldsymbol{\beta}_1,\boldsymbol{\beta}_2\left|\boldsymbol{\beta}_0\right.\right)$ sum of squares of regression for the model in Problem 10.12 = 35092.6

$$SS_R\left(\boldsymbol{\beta}_1\left|\boldsymbol{\beta}_0\right.\right)=34502.3$$

$$SS_R\left(\boldsymbol{\beta}_2\left|\boldsymbol{\beta}_{0,}\boldsymbol{\beta}_1\right.\right)=35092.6-34502.3=590.3$$

$$F_0=\frac{SS_R\left(\boldsymbol{\beta}_2\left|\boldsymbol{\beta}_{0,}\boldsymbol{\beta}_1\right.\right)/3}{MS_E}=\frac{590.3/3}{36.511}=5.3892$$

Since $F_{0.05,3,6} = 4.76$, then the addition of the quadratic terms to the model is significant. The *p*-values indicate that it's probably the term x_2^2 that is responsible for this.

10.11. ***Relationship between analysis of variance and regression.*** Any analysis of variance model can be expressed in terms of the general linear model $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$, where the $\mathbf{X}$ matrix consists of zeros and ones. Show that the single-factor model $y_{ij} = \mu + \tau_i + \varepsilon_{ij}$, i=1,2,3, j=1,2,3,4 can be written in general linear model form. Then

(a) Write the normal equations $(\mathbf{X}'\mathbf{X})\hat{\boldsymbol{\beta}} = \mathbf{X}'\mathbf{y}$ and compare them with the normal equations found for the model in Chapter 3.

The normal equations are $(\mathbf{X}'\mathbf{X})\hat{\boldsymbol{\beta}} = \mathbf{X}'\mathbf{y}$

$$\begin{bmatrix} 12 & 4 & 4 & 4 \\ 4 & 4 & 0 & 0 \\ 4 & 0 & 4 & 0 \\ 4 & 0 & 0 & 4 \end{bmatrix} \begin{bmatrix} \hat{\mu} \\ \hat{\tau}_1 \\ \hat{\tau}_2 \\ \hat{\tau}_3 \end{bmatrix} = \begin{bmatrix} y_{..} \\ y_{1.} \\ y_{2.} \\ y_{3.} \end{bmatrix}$$

which are in agreement with the results of Chapter 3.

(b) Find the rank of $\mathbf{X}'\mathbf{X}$. Can $(\mathbf{X}'\mathbf{X})^{-1}$ be obtained?

$\mathbf{X}'\mathbf{X}$ is a 4 x 4 matrix of rank 3, because the last three columns add to the first column. Thus $(\mathbf{X}'\mathbf{X})^{-1}$ does not exist.

(c) Suppose the first normal equation is deleted and the restriction $\sum_{i=1}^{3} n\hat{\tau}_i = 0$ is added. Can the resulting system of equations be solved? If so, find the solution. Find the regression sum of squares $\hat{\boldsymbol{\beta}}'\mathbf{X}'\mathbf{y}$, and compare it to the treatment sum of squares in the single-factor model.

Imposing $\sum_{i=1}^{3} n\hat{\tau}_i = 0$ yields the normal equations

$$\begin{bmatrix} 0 & 4 & 4 & 4 \\ 4 & 4 & 0 & 0 \\ 4 & 0 & 4 & 0 \\ 4 & 0 & 0 & 4 \end{bmatrix} \begin{bmatrix} \hat{\mu} \\ \hat{\tau}_1 \\ \hat{\tau}_2 \\ \hat{\tau}_3 \end{bmatrix} = \begin{bmatrix} y_{..} \\ y_{1.} \\ y_{2.} \\ y_{3.} \end{bmatrix}$$

The solution to this set of equations is

$$\hat{\mu} = \frac{y_{..}}{12} = \bar{y}_{..}$$
$$\hat{\tau}_i = \bar{y}_{i.} - \bar{y}_{..}$$

This solution was found by solving the last three equations for $\hat{\tau}_i$, yielding $\hat{\tau}_i = \bar{y}_{i.} - \hat{\mu}$, and then substituting in the first equation to find $\hat{\mu} = \bar{y}_{..}$

The regression sum of squares is

$$SS_R(\boldsymbol{\beta}) = \hat{\boldsymbol{\beta}}' = \mathbf{X}'\mathbf{y} = \bar{y}_{..}y_{..} + \sum_{i=1}^{a}\left(\bar{y}_{i.} - y_{..}\right)^2 = \frac{y_{..}^2}{an} + \sum_{i=1}^{a}\frac{\bar{y}_{i.}^2}{n} - \frac{y_{..}^2}{an} = \sum_{i=1}^{a}\frac{\bar{y}_{i.}^2}{n}$$

with *a* degrees of freedom. This is the same result found in Chapter 3. For more discussion of the relationship between analysis of variance and regression, see Montgomery and Peck (1992).

10.12. Suppose that we are fitting a straight line and we desire to make the variance of $\hat{\beta}_1$ as small as possible. Restricting ourselves to an even number of experimental points, where should we place these points so as to minimize $V\left(\hat{\beta}_1\right)$? (Note: Use the design called for in this exercise with great caution because, even though it minimizes $V\left(\hat{\beta}_1\right)$, it has some undesirable properties; for example, see Myers and Montgomery (1995). Only if you are *very sure* the true functional relationship is linear should you consider using this design.

Since $V\left(\hat{\beta}_1\right) = \frac{\sigma^2}{S_{xx}}$, we may minimize $V\left(\hat{\beta}_1\right)$ by making S_{xx} as large as possible. S_{xx} is maximized by spreading out the x_j's as much as possible. The experimenter usually has a "region of interest" for *x*. If *n* is even, *n*/2 of the observations should be run at each end of the "region of interest". If *n* is odd, then run one of the observations in the center of the region and the remaining (*n*-1)/2 at either end.

10.13. ***Weighted least squares.*** Suppose that we are fitting the straight line $y = \beta_0 + \beta_1 x + \varepsilon$, but the variance of the *y*'s now depends on the level of *x*; that is,

$$V\left(y|x_i\right) = \sigma_i^2 = \frac{\sigma^2}{w_i}, i = 1, 2, ..., n$$

where the w_i are known constants, often called weights. Show that if we choose estimates of the regression coefficients to minimize the weighted sum of squared errors given by $\sum_{i=1}^{n} w_i\left(y_i - \beta_0 + \beta_1 x_i\right)^2$, the resulting least squares normal equations are

$$\hat{\beta}_0\sum_{i=1}^{n} w_i + \hat{\beta}_1\sum_{i=1}^{n} w_i x_i = \sum_{i=1}^{n} w_i y_i$$

$$\hat{\beta}_0\sum_{i=1}^{n} w_i x_i + \hat{\beta}_1\sum_{i=1}^{n} w_i x_i^2 = \sum_{i=1}^{n} w_i x_i y_i$$

The least squares normal equations are found:

$$L = \sum_{i=1}^{n}\left(y_i - \beta_0 - \beta_1 x_1\right)^2 w_i$$

$$\frac{\partial L}{\partial \beta_0} = -2\sum_{i=1}^{n}\left(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_1\right)w_i = 0$$

$$\frac{\partial L}{\partial \beta_1} = -2\sum_{i=1}^{n}\left(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_1\right)x_1 w_i = 0$$

which simplify to

$$\hat{\beta}_0 \sum_{i=1}^{n} w_i + \hat{\beta}_1 \sum_{i=1}^{n} x_1 w_i = \sum_{i=1}^{n} w_i y_i$$
$$\hat{\beta}_0 \sum_{i=1}^{n} x_1 w_i + \hat{\beta}_1 \sum_{i=1}^{n} x_1^2 w_i = \sum_{i=1}^{n} w_i x_1 y_i$$

10.14. Consider the 2_{III}^{7-4}. Suppose after running the experiment, the largest observed effects are *A* + *BD*, *B* + *AD*, and *D* + *AB*. You wish to augment the original design with a group of four runs to de-alias these effects.

(a) Which four runs would you make?

Take the first four runs of the original experiment and change the sign on *A*.

Design Expert Output

Std	Run	Block	Factor 1 A:x1	Factor 2 B:x2	Factor 3 C:x3	Factor 4 D:x4	Factor 5 E:x5	Factor 6 F:x6	Factor 7 G:x7
1	1	Block 1	-1	-1	-1	1	1	1	-1
2	2	Block 1	1	-1	-1	-1	-1	1	1
3	3	Block 1	-1	1	-1	-1	1	-1	1
4	4	Block 1	1	1	-1	1	-1	-1	-1
5	5	Block 1	-1	-1	1	1	-1	-1	1
6	6	Block 1	1	-1	1	-1	1	-1	-1
7	7	Block 1	-1	1	1	-1	-1	1	-1
8	8	Block 1	1	1	1	1	1	1	1
9	9	Block 2	1	-1	-1	1	1	1	-1
10	10	Block 2	-1	-1	-1	-1	-1	1	1
11	11	Block 2	1	1	-1	-1	1	-1	1
12	12	Block 2	-1	1	-1	1	-1	-1	-1

Main effects and interactions of interest are:

x1	x2	x4	x1x2	x1x4	x2x4
-1	-1	1	1	-1	-1
1	-1	-1	-1	-1	1
-1	1	-1	-1	1	-1
1	1	1	1	1	1
-1	-1	1	1	-1	-1
1	-1	-1	-1	-1	1
-1	1	-1	-1	1	-1
1	1	1	1	1	1
1	**-1**	**1**	**-1**	**1**	**-1**
-1	**-1**	**-1**	**1**	**1**	**1**
1	**1**	**-1**	**1**	**-1**	**-1**
-1	**1**	**1**	**-1**	**-1**	**1**

(b) Find the variances and covariances of the regression coefficients in the model

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_4 x_4 + \beta_{12} x_1 x_2 + \beta_{14} x_1 x_4 + \beta_{24} x_2 x_4 + \varepsilon$$

$$\mathbf{X}'\mathbf{X} = \begin{bmatrix} 12 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 12 & 0 & 0 & 0 & 0 & 4 \\ 0 & 0 & 12 & 0 & 0 & 4 & 0 \\ 0 & 0 & 0 & 12 & 4 & 0 & 0 \\ 0 & 0 & 0 & 4 & 12 & 0 & 0 \\ 0 & 0 & 4 & 0 & 0 & 12 & 0 \\ 0 & 4 & 0 & 0 & 0 & 0 & 12 \end{bmatrix}$$

$$\left(\mathbf{X}'\mathbf{X}\right)^{-1} = \begin{bmatrix} 0.08333 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0.09375 & 0 & 0 & 0 & 0 & -0.03125 \\ 0 & 0 & 0.09375 & 0 & 0 & -0.03125 & 0 \\ 0 & 0 & 0 & 0.09375 & -0.03125 & 0 & 0 \\ 0 & 0 & 0 & -0.03125 & 0.09375 & 0 & 0 \\ 0 & 0 & -0.03125 & 0 & 0 & 0.09375 & 0 \\ 0 & -0.03125 & 0 & 0 & 0 & 0 & 0.09375 \end{bmatrix}$$

(c) Is it possible to de-alias these effects with fewer than four additional runs?

It is possible to de-alias these effects in only two runs. By utilizing *Design Expert's* design augmentation *D*-optimal factorial function, the runs 9 and 10 (Block 2) were generated as follows:

Design Expert Output

Std	Run	Block	Factor 1 A:x1	Factor 2 B:x2	Factor 3 C:x3	Factor 4 D:x4	Factor 5 E:x5	Factor 6 F:x6	Factor 7 G:x7
1	1	Block 1	-1	-1	-1	1	1	1	-1
2	2	Block 1	1	-1	-1	-1	-1	1	1
3	3	Block 1	-1	1	-1	-1	1	-1	1
4	4	Block 1	1	1	-1	1	-1	-1	-1
5	5	Block 1	-1	-1	1	1	-1	-1	1
6	6	Block 1	1	-1	1	-1	1	-1	-1
7	7	Block 1	-1	1	1	-1	-1	1	-1
8	8	Block 1	1	1	1	1	1	1	1
9	9	Block 2	1	-1	-1	1	-1	-1	-1
10	10	Block 2	-1	-1	-1	-1	-1	-1	-1

10.15. Consider the computer output from a regression model shown below.

Source	Sum of Squares	Degrees of Freedom	Mean Square	*F*
Model	532,567.14	2	?	?
Error	?	?	?	
Total	539,534.33	14		

(a) Fill in the blanks in the table.

Source	Sum of Squares	Degrees of Freedom	Mean Square	*F*
Model	532,567.14	2	266,283.57	458.64
Error	6,967.12	12	580.60	
Total	539,534.33	14		

(b) How many regressor or predictor variables are in this model?

2 regressors.

(c) Find bounds on the *P*-value for the test on significance of regression.

0.0000

(d) Suppose that the *t*-statistic for one of the predictor variables in this model is 4.22. Find bounds on the *P*-value.

Minitab Output

Probability Density Function

```
Student's t distribution with 12 DF

   x     f( x )
4.22  0.0010552
```

10.16. Consider the computer output from a regression model shown below.

Source	Sum of Squares	Degrees of Freedom	Mean Square	*F*
Model	534.66	?	133.67	?
Error	?	?	?	
Total	590.25	19		

(a) Fill in the blanks in the table.

Source	Sum of Squares	Degrees of Freedom	Mean Square	*F*
Model	534.66	4	133.67	36.07
Error	55.59	15	3.71	
Total	590.25	19		

(b) How many regressor or predictor variables are in this model?

4 regressors.

(c) Compute the *F*-statistic for significance of regression and find the bounds on the *P*-value. 0.0000

0.0000

(d) Suppose that the *t*-statistic for one of the predictor variables in this model is 6.11. Find bounds on the *P*-value.

Minitab Output

Probability Density Function

```
Student's t distribution with 15 DF

   x     f( x )
6.11  0.0000179
```

(e) What is the value of the R^2 statistic for this model?

$$R^2 = \frac{SS_R}{SS_T} = 1 - \frac{SS_E}{SS_T} = 1 - \frac{55.59}{590.25} = 0.906 \text{ or } 90.6\%$$

10.17. A regression model with 5 regressors has been built to 75 observations. The individual *t*-statistics for each predictor are as follows: 4.21, 2.12, 6.98, 3.55, and 2.45.

(a) Find the bounds on the *P*-values for each predictor.

Minitab Output

Probability Density Function
Student's t distribution with 69 DF

x	f(x)
4.21	0.0001331

Probability Density Function
Student's t distribution with 69 DF

x	f(x)
2.12	0.0436673

Probability Density Function
Student's t distribution with 69 DF

x	f(x)
6.98	0.0000000

Probability Density Function
Student's t distribution with 69 DF

x	f(x)
3.55	0.0011206

Probability Density Function
Student's t distribution with 69 DF

x	f(x)
2.45	0.0214488

(b) Based on your answers to part (a), are all of the regressor variables currently in the model necessary?

Yes.

10.18. A regression model with 3 regressors has been fit to a sample of 45 observations. The total sum of squares for the model is 275.6 and the model sum of squares is 245.86.

(a) Find the value of the mean square error.

Source	Sum of Squares	Degrees of Freedom	Mean Square	*F*
Model	245.86	3	81.95	112.98
Error	29.74	41	0.725	
Total	275.60	45-1=44		

(b) Find the model mean square. 81.95

(c) Compute the value of the *F*-statistic for significance of regression and find bounds on the *P*-value. What are your conclusions? P=0.0000

(d) What is the value of R^2 for this model?

$$R^2 = \frac{SS_R}{SS_T} = 1 - \frac{SS_E}{SS_T} = 1 - \frac{29.74}{275.6} = 0.892 \text{ or } 89.2\%$$

(e) What is the value of the adjusted R^2 statistic for this model?

$$R^2_{adj} = 1 - \left(\frac{n-1}{n-p}\right)\left(1-R^2\right) = 1 - \left(\frac{45-1}{45-3}\right)(1-0.892) = 1 - \left(\frac{44}{42}\right)(0.1079) = 0.887 \text{ or } 88.7\%$$

10.19. A regression model with 4 regressors has been fit to a sample of 65 observations. The total sum of squares for the model is 300 and the model sum of squares is 280.

(a) What is the value of R^2 for this model?

$$R^2 = \frac{SS_R}{SS_T} = 1 - \frac{SS_E}{SS_T} = 1 - \frac{300-280}{300} = 0.933 \text{ or } 93.3\%$$

(b) What is the value of the adjusted R^2 statistic for this model?

$$R^2_{adj} = 1 - \left(\frac{n-1}{n-p}\right)\left(1-R^2\right) = 1 - \left(\frac{65-1}{65-3}\right)(1-0.933) = 1 - \left(\frac{64}{62}\right)(0.06667) = 0.931 \text{ or } 93.1\%$$

(c) Suppose that one of the regressors is removed from the model and the new model sum of squares is 250. What is the value of the adjusted R^2 for this reduced model? What is the impact on model fit of removing this regressor?

$$R^2 = \frac{SS_R}{SS_T} = 1 - \frac{SS_E}{SS_T} = 1 - \frac{300-250}{300} = 0.833 \text{ or } 83.3\%$$

$$R^2_{adj} = 1 - \left(\frac{n-1}{n-p}\right)\left(1-R^2\right) = 1 - \left(\frac{65-1}{65-2}\right)(1-0.833) = 1 - \left(\frac{64}{63}\right)(0.16667) = 0.828 \text{ or } 82.8\%$$

Removing the regressor reduced both R^2 values. Recommend the model with 3 regressors.

10.20. A regression model with 3 regressors has been built to a sample of 30 observations. The individual *t*-statistics for each predictor are as follows: 2.12, 5.45, and 17.50.

(a) Find bounds on the *P*-values for each predictor.

Minitab Output

Probability Density Function

```
Student's t distribution with 26 DF
   x     f( x )
2.12  0.0459097
```

Probability Density Function

```
Student's t distribution with 26 DF
   x     f( x )
5.45  0.0000135
```

Probability Density Function

```
Student's t distribution with 26 DF
   x     f( x )
17.5  0.0000000
```

(b) Based on your answers to part (a), are all of the regressor variables currently in the model necessary?

Yes.

10.21. The value of the adjusted R^2 statistic always increases when a new regressor variable is added to the model.

True **False**

10.22. The value of the ordinary R^2 statistic can decrease when a new regressor variable is added to the model.

True **False**

10.23. The ordinary R^2 statistic is a good indicator of the prediction capability of a regression model.

True **False**

10.24. If the P-value for the test for significance of regression is <0.01, this is an indication that all of the regressor variables in the model are necessary.

True **False**

10.25. If the *P*-value for the test for significance of regression is <0.01, this is an indication that the regression model will be a good predictor of new observations.

True **False**

10.26. A small value of the PRESS statistic is a good indication that the regression model will be a good predictor of new observations.

True False

10.27. The *t*-test on individual regression coefficients is a test on the utility of the regressor as if it were the last regressor added to a model that already contains the other regressors.

True False